Q1 - 2024 (01 Feb Shift 1)

A simple pendulum of length 1 m has a wooden bob of mass 1 kg. It is struck by a bullet of mass 10^{-2} kg moving with a speed of $2 \times 10^2 \mathrm{~ms}^{-1}$. The bullet gets embedded into the bob. The height to which the bob rises before swinging back is. (use $\mathrm{g}=10 \mathrm{~m/s}^2$)

(1) 0.30 m

(2) 0.20 m

(3) 0.35 m

(4) 0.40 m

Q2 - 2024 (01 Feb Shift 1)

The identical spheres each of mass 2M are placed at the corners of a right angled triangle with mutually perpendicular sides equal to 4 m each. Taking point of intersection of these two sides as origin, the magnitude of position vector of the centre of mass of the system is $\frac{4\sqrt{2}}{x}$, where the value of x is_____

Q3 - 2024 (01 Feb Shift 2)

A uniform rod AB of mass 2 kg and Length 30 cm at rest on a smooth horizontal surface. An impulse of force 0.2Ns is applied to end B. The time taken by the rod to turn through at right angles will be $\frac{\pi}{\mathrm{x}}$s, where $\mathrm{x}=$ _____

Q4 - 2024 (30 Jan Shift 1)

A spherical body of mass 100 g is dropped from a height of 10 m from the ground. After hitting the ground, the body rebounds to a height of 5 m. The impulse of force imparted by the ground to the body is given by: (given $\mathrm{g}=9.8 \mathrm{~m/s}^2$)

(1) $4.32 \mathrm{~kg~ms}^{-1}$

(2) $43.2 \mathrm{~kg~ms}^{-1}$

(3) $23.9 \mathrm{~kg~ms}^{-1}$

(4) $2.39 \mathrm{~kg~ms}^{-1}$

Q5 - 2024 (30 Jan Shift 2)

If the total energy transferred to a surface in time t is 6.48×10^5 J, then the magnitude of the total momentum delivered to this surface for complete absorption will be :

(1) 2.46×10^{-3} kg m/s

(2) 2.16×10^{-3} kg m/s

(3) 1.58×10^{-3} kg m/s

(4) 4.32×10^{-3} kg m/s

Q6 - 2024 (31 Jan Shift 1)

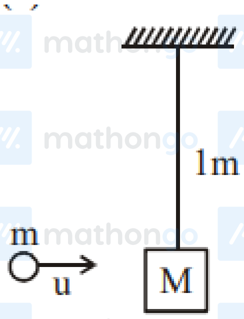
A solid circular disc of mass 50 kg rolls along a horizontal floor so that its center of mass has a speed of 0.4 m/s. The absolute value of work done on the disc to stop it is_____ J.

Answer Key

Q1 (2) **Q2** (3) **Q3** (4) **Q4** (4)

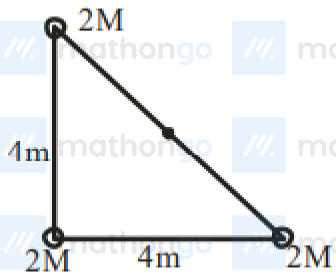
Q5 (2) **Q6** (6)

Q1



$\mathrm{mu} = (\mathrm{M} + \mathrm{m})\mathrm{V}$
$10^{-2} \times 2 \times 10^{2} \cong 1 \times \mathrm{V}$
$\mathrm{V} \cong 2\ \mathrm{m/s}$
$\mathrm{h} = \dfrac{\mathrm{V}^2}{2\ \mathrm{g}} = 0.2\ \mathrm{m}$

Q2



Position vector $\vec{r}_{\mathrm{COM}} = \dfrac{\mathrm{m}_1 \vec{r}_1 + \mathrm{m}_2 \vec{r}_2 + \mathrm{m}_3 \vec{r}_3}{\mathrm{m}_1 + \mathrm{m}_2 + \mathrm{m}_3}$

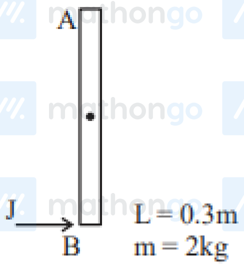
$\vec{r}_{\mathrm{COM}} = \dfrac{2\mathrm{M} \times 0 + 2\mathrm{M} \times 4\hat{i} + 2\mathrm{M} \times 4\hat{j}}{6\mathrm{M}}$

$\vec{r} = \dfrac{4}{3}\hat{i} + \dfrac{4}{3}\hat{j}$

$|\vec{r}| = \dfrac{4\sqrt{2}}{3}$

$x = 3$

Q3



Impulse $J = 0.2\ \mathrm{N-S}$

$J = \int \mathrm{Fdt} = 0.2\ \mathrm{N-s}$

Angular impuls $(\overrightarrow{\mathrm{M}})$

$$\overrightarrow{\mathrm{M}}_{\mathrm{c}} = \int \tau \mathrm{dt}$$

$$= \int \mathrm{F}\frac{\mathrm{L}}{2}\mathrm{dt}$$

$$= \frac{\mathrm{L}}{2}\int \mathrm{Fdt} = \frac{\mathrm{L}}{2} \times \mathrm{J}$$

$$= \frac{0.3}{2} \times 0.2$$

$$= 0.03$$

$$I_{\mathrm{cm}} = \frac{\mathrm{L}^2}{12} = \frac{2 \times (0.3)^2}{12} = \frac{0.09}{6}$$

$$\mathrm{M} = \mathrm{I}_{\mathrm{cm}}(\omega_{\mathrm{f}} - \omega_{\mathrm{i}})$$

$$0.03 = \frac{0.09}{6}(\omega_{\mathrm{f}})$$

$$\omega_{\mathrm{f}} = 2\mathrm{rad/s}$$

$$\theta = \omega \mathrm{t}$$

$$\mathrm{t} = \frac{\theta}{\omega} = \frac{\pi}{2 \times 2} = \frac{\pi}{4}\mathrm{sec.}$$

$$X = 4$$

Q4

$$\vec{I} = \Delta\vec{P} = \vec{P}_f - \vec{P}_i$$

$$\mathrm{M} = 0.1\ \mathrm{kg}$$

$$I = \Delta P = 0.1(\sqrt{2 \times 9.8 \times 5} - (-\sqrt{2 \times 9.8 \times 10}))$$

$$= 0.1(14 + 7\sqrt{2}) \approx 2.39\ \mathrm{kg\ ms^{-1}}$$

Q5

$$p = \frac{E}{C} = \frac{6.48\times10^5}{3\times10^8} = 2.16\times10^{-3}$$

Q6

Using work energy theorem

$$W = \Delta KE = 0 - \left(\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2\right)$$

$$W = 0 - \frac{1}{2}mv^2\left(1 + \frac{K^2}{R^2}\right)$$

$$= -\frac{1}{2}\times 50\times 0.4^2\left(1+\frac{1}{2}\right) = -6\ J$$

Absolute work $= +6$ J

$W = -6J \quad |W| = 6J$